

- Q.30 Differentiate between online and Off-line UPS.
(CO7)
- Q.31 What do you mean by an electric drive? Differentiate between ac drives and dc drives.
(CO6)
- Q.32 Explain the working of step-down chopper with circuit diagram and voltage current wave forms.
- Q.33 Explain the working of single phase half wave controlled rectifier with resistive load showing input-output current voltage waveforms.
- Q.34 Draw the block diagram of fixed frequency variable voltage control of a single phase induction motor.
(CO6)
- Q.35 Explain how an SCR can be used for light intensity control.
(CO1)

SECTION-D

- Note:** Long answer type questions. Attempt any two questions out of three questions. (2x10=20)
- Q.36 Explain the construction, principle of working and VI-characteristics SCR with neat circuit diagram and wave form.
(CO1)
- Q.37 Draw the circuit diagram of d.c. chopper scheme for speed control of a d.c. shunt motor and explain its working.
(CO6)
- Q.38 Why a series inverter is called so? Which type of commutation is used in a basic series inverter? Explain its working principle with the aid of waveforms.(CO4)

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6th Sem / Electrical Engg.

Sub : Industrial Electronics & Control of Drives

Time : 3Hrs.

M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 A thyristor (SCR) is a (CO1)
- Current controlled device
 - Bilateral device
 - Voltage controlled device
 - None of the above
- Q.2 A TRIAC is a _____ switch (CO2)
- Bidirectional
 - Unidirectional
 - Tri directional
 - None of the above
- Q.3 A UJT is a (CO2)
- Current controlled device
 - A relaxation oscillator
 - Voltage controlled device
 - None of the above
- Q.4 The output voltage of a controlled rectifier is maximum when firing angle is (CO3)
- 120°
 - 60°
 - 0°
 - 90°
- Q.5 Find the output voltage for a step up chopper when it is operated at a duty cycle of 50% and $V_s = 240V$. (CO4)
- 240 V
 - 480 V
 - 560 V
 - 120 V

- Q.6 The commutation method in an inverter is (CO5)
 a) Line commutation b) Forced commutation
 c) Either (a) or (b) d) None of the above
- Q.7 A step down choppers can be used in (CO5)
 a) Electric traction b) Electric vehicles
 c) Machine tools d) All of these
- Q.8 _____ can be used as a dc static switch. (CO6)
 a) GTO
 b) Transistor
 c) Both GTO & Transistor
 d) TRIAC
- Q.9 Field control method is used in D.C. motor to get. (CO6)
 a) Speed above normal speed
 b) Speeds below normal speed
 c) Both A & B
 d) None of above
- Q.10 UPS is never used in (CO7)
 a) Street lighting b) Communication link
 c) Computer d) All of the above

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

- Q.11 While conducting, the resistance of SCR is _____. (CO1)
- Q.12 Speed of a motor _____ with increases in firing angle. (CO6)
- Q.13 In an SCR, Latching current is more than holding current (True/False)
- Q.14 An SCR is a _____ layer _____ junction device. (CO1)

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- Q.15 Full wave rectifiers allow power flow in both directions. (True/False) (CO3)
- Q.16 A switching mode power supply is a multistage _____ Supply (CO7)
- Q.17 The duty cycle of a chopper is expressed as _____. (Co6)
- Q.18 DC motors provide _____ starting torque. (CO6)
- Q.19 The main application of a cyclo - converter circuits is found in _____. (CO5)
- Q.20 SMF batteries are costier than lead-acid batteries. (True/False) (CO7)

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

- Q.21 Differentiate between SCR and TRIAC. (CO2)
- Q.22 Draw the V-I characteristic of SCR and explain its working. (CO1)
- Q.23 Draw and explain circuit & waveform of single phase half controlled rectifier with resistive load. (CO3)
- Q.24 Why is bridge circuit preferred than the centre tapped transformer circuit? (CO3)
- Q.25 Explain the series inverter and mention their important applications. (CO4)
- Q.26 What are the different types of commutation. (CO2)
- Q.27 Compare two modes of dual converter. (CO5)
- Q.28 What is cyclo converter? Explain the single phase to single phase step up cyclo converter. (CO5)
- Q.29 Draw and explain the circuit diagram of an automatic battery charger. (CO6)

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